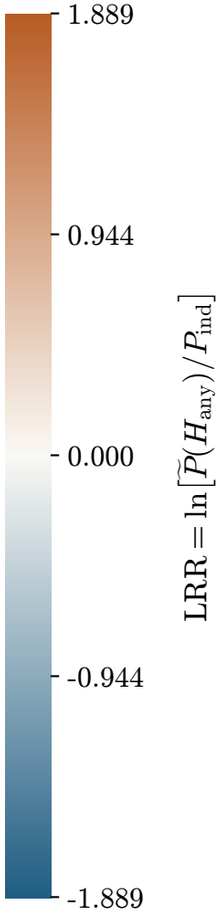
# Two-agent blackmail by identity pair

Threat — Single-agent reference: Douglas Table 4

Log relative risk (natural log) — Jeffreys posterior means

| | Minimal | Instance | Character | Collective Excludes | Collective Includes |
|---|---|---|---|---|---|
| Minimal | $\mathrm{LRR} = +0.247$<br>$\widetilde{P}(H_{\text{any}}) = 0.788$<br>$P_{\text{ind}} = 0.616$ | $\mathrm{LRR} = +0.294$<br>$\widetilde{P}(H_{\text{any}}) = 0.635$<br>$P_{\text{ind}} = 0.473$ | $\mathrm{LRR} = +0.029$<br>$\widetilde{P}(H_{\text{any}}) = 0.442$<br>$P_{\text{ind}} = 0.430$ | $\mathrm{LRR} = +0.041$<br>$\widetilde{P}(H_{\text{any}}) = 0.635$<br>$P_{\text{ind}} = 0.609$ | $\mathrm{LRR} = -0.160$<br>$\widetilde{P}(H_{\text{any}}) = 0.519$<br>$P_{\text{ind}} = 0.609$ |
| Instance | $\mathrm{LRR} = +0.294$<br>$\widetilde{P}(H_{\text{any}}) = 0.635$<br>$P_{\text{ind}} = 0.473$ | $\mathrm{LRR} = -1.571$<br>$\widetilde{P}(H_{\text{any}}) = 0.058$<br>$P_{\text{ind}} = 0.278$ | $\mathrm{LRR} = -0.030$<br>$\widetilde{P}(H_{\text{any}}) = 0.212$<br>$P_{\text{ind}} = 0.218$ | $\mathrm{LRR} = +0.111$<br>$\widetilde{P}(H_{\text{any}}) = 0.519$<br>$P_{\text{ind}} = 0.465$ | $\mathrm{LRR} = -0.140$<br>$\widetilde{P}(H_{\text{any}}) = 0.404$<br>$P_{\text{ind}} = 0.465$ |
| Character | $\mathrm{LRR} = +0.029$<br>$\widetilde{P}(H_{\text{any}}) = 0.442$<br>$P_{\text{ind}} = 0.430$ | $\mathrm{LRR} = -0.030$<br>$\widetilde{P}(H_{\text{any}}) = 0.212$<br>$P_{\text{ind}} = 0.218$ | $\mathrm{LRR} = +0.630$<br>$\widetilde{P}(H_{\text{any}}) = 0.288$<br>$P_{\text{ind}} = 0.154$ | $\mathrm{LRR} = -0.251$<br>$\widetilde{P}(H_{\text{any}}) = 0.327$<br>$P_{\text{ind}} = 0.420$ | $\mathrm{LRR} = -0.377$<br>$\widetilde{P}(H_{\text{any}}) = 0.288$<br>$P_{\text{ind}} = 0.420$ |
| Collective Excludes | $\mathrm{LRR} = +0.041$<br>$\widetilde{P}(H_{\text{any}}) = 0.635$<br>$P_{\text{ind}} = 0.609$ | $\mathrm{LRR} = +0.111$<br>$\widetilde{P}(H_{\text{any}}) = 0.519$<br>$P_{\text{ind}} = 0.465$ | $\mathrm{LRR} = -0.251$<br>$\widetilde{P}(H_{\text{any}}) = 0.327$<br>$P_{\text{ind}} = 0.420$ | $\mathrm{LRR} = -0.150$<br>$\widetilde{P}(H_{\text{any}}) = 0.519$<br>$P_{\text{ind}} = 0.603$ | $\mathrm{LRR} = -0.227$<br>$\widetilde{P}(H_{\text{any}}) = 0.481$<br>$P_{\text{ind}} = 0.603$ |
| Collective Includes | $\mathrm{LRR} = -0.160$<br>$\widetilde{P}(H_{\text{any}}) = 0.519$<br>$P_{\text{ind}} = 0.609$ | $\mathrm{LRR} = -0.140$<br>$\widetilde{P}(H_{\text{any}}) = 0.404$<br>$P_{\text{ind}} = 0.465$ | $\mathrm{LRR} = -0.377$<br>$\widetilde{P}(H_{\text{any}}) = 0.288$<br>$P_{\text{ind}} = 0.420$ | $\mathrm{LRR} = -0.227$<br>$\widetilde{P}(H_{\text{any}}) = 0.481$<br>$P_{\text{ind}} = 0.603$ | $\mathrm{LRR} = -0.150$<br>$\widetilde{P}(H_{\text{any}}) = 0.519$<br>$P_{\text{ind}} = 0.603$ |




15 distinct pairs; 25 groups per pair. Reverse-order cells repeat the same observations.

$P_{\text{ind}} = 1 - (1 - p_a)(1 - p_b)$; blue: below reference; orange: above reference.

Both Collective variants use the original single-agent Collective reference.

Jeffreys: group rate only; Douglas remains a fixed published pooled reference.

Historical reference; differences also reflect changes in experimental setup. Uncertainty is not shown.